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The United Nations World Water Development Report 2024

Water for prosperity and peace

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SHORT SUMMARY

Water security leads to prosperity and peace, while the hardships of conflict are amplified through water

Water nurtures prosperity by meeting basic human needs, supporting health, livelihoods and economic development, underpinning food and energy security, and defending environmental integrity.

Water influences the economy in many ways, and global trade dynamics and market adaptations can have direct repercussions on the water use of regional and local economies. The water-related impacts of conflict are multi-faceted and often indirect, such as those linked to forced migration and increased exposure to health threats.

Water directly supports billions of livelihoods and can promote peace

Climate change, geopolitical unrest, pandemics, mass migration, hyperinflation and other crises can exacerbate water access inequalities. In nearly all cases, the poorest and most vulnerable groups are those that suffer the greatest risks to their well-being.

The ***2024 edition*** of the ***United Nations World Water Development Report (UN WWDR)*** calls attention to the complex and interlinked relationships between water, prosperity and peace, describing how progress in one dimension can have positive, often essential, repercussions on the others.

"Since wars begin in the minds of men and women it is in the minds of men and women that the defences of peace must be constructed"

The United Nations World Water Development Report 2024

Water for prosperity and peace

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Chapter 10

Science, technology and information

WWAP
Matthew England and Richard Connor

With contribution from: Tommaso Abrate (WMO)

A central pillar of informing better technical and management decisions is the availability of accurate data and information (UNESCO/UN-Water, 2020). Advances in science and technology have generated an unprecedented volume of data and information regarding the state of water resources and the operational effects of management interventions at the global, regional, national, river basin and field level. Where available and when accessible, this increased level of knowledge (i.e. data and information) has been used to inform and improve policy development, operational water management decisions and technical interventions.

Real-time data and information covering relatively short timescales are particularly useful for operational decisions

Real-time data and information covering relatively short timescales (e.g. minute to hour) are particularly useful for operational decisions such as early warning systems, and for managing infrastructure to mitigate flood risk. Similarly, mid-to-long-term data (e.g. intra- and inter-annual) have provided insights to support the strategic design of water infrastructure and scenario-based planning. Such advances have in turn promoted prosperity and peace within societies through the expansion of the irrigation area, the increase in agricultural production as exemplified by the Green Revolution, better access to drinking water and sanitation, improved disaster risk reduction, and the introduction of water-efficient industrial processes.

However, there still exists a significant lack of historical and up-to-date data and information on surface and groundwater, soil moisture, and associated hydro-meteorological parameters. Furthermore, historical (time-series) data become less reliable due to increasing climate variability (and change), posing challenges to the planning and design of water infrastructure (IPCC, 2022; Milly et al., 2008). This hinders progress towards Sustainable Development Goals (SDGs), particularly the targets of SDG 6, on water, and also other goals such as SDG 2 on zero hunger. Where data and information do exist, forthcoming and transparent sharing between user groups can be problematic in some cases, particularly in the case of transboundary data between riparians (United Nations, 2023).

The SDG 6 Global Acceleration Framework identifies data and information as a key component to build trust through data generation, validation, standardization and information exchange for decision-making and accountability in water management (UN-Water, 2020). Progress towards SDG 6 targets is only comprehensively reported for drinking water and sanitation, with only rough indications of progress for indicators such as water stress, water use efficiency, transboundary cooperation and integrated water resources management (IWRM), leaving 5 of the 11 indicators with little (if any) quantified information on progress. These data gaps are largely due to monitoring and reporting deficiencies (United Nations, 2023). In 2023, the UN-Water Integrated Monitoring Initiative for SDG 6 initiated a renewed data drive, requesting data and offering associated capacity-building.

10.1 Science, technology and innovation

Advances in science have played a critical role in technological development and innovation within the water sector. This has led to the development of innovative tools and approaches to measure and monitor parameters of the hydrosphere. These include, but are not limited to, information and communication technologies; earth observation and space technology through the deployment of satellites and remote sensing; advanced sensor technology to monitor hydrological systems; the rise of citizen science supported by low-cost technologies; and the application of 'big data' analytics (UNESCO/UN-Water, 2020).

Artificial intelligence (AI) has been proposed to help address challenges across water supply, sanitation and hygiene (WASH) systems, water use in agriculture and industry, and water resources management. It is claimed that AI has the ability to enhance supply insights into catchment management, emergency response, wastewater treatment plants and distribution networks, operations and maintenance, and demand management (Richards et al., 2023). However, the performance of any AI tool also requires data. The benefits of AI are caveated with the warning that the impacts of this nascent technology remain largely unknown, with the potential to trigger serious and unexpected problems. These include system-wide compromise owing to design errors, malfunction and cyber-attacks (Box 10.1), which in turn could lead to critical infrastructure failure in a worst-case scenario. Mitigation strategies to alleviate risks associated with AI include addressing gaps in foundational infrastructure and digital literacy; establishing institutional, software and hardware mechanisms for trustworthy AI; and developing detailed risk–benefit analyses (Richards et al., 2023).

Water consumption by information technology companies[49] has significantly increased in recent years, by up to a third. A major share of this is attributed to the development of AI and related technologies. Large volumes of water are used in the liquid cooling systems of computers that run AI programmes, in addition to the energy required to power the equipment. It is estimated that AI currently requires 500 ml of water to answer 10–50 queries,[50] dependent on the weather and season, and energy generation water use efficiency. The simulated training of GPT-3 in state-of-the-art data centres in the United States of America (USA) consumes and estimated 700,000 l of water (Li et al., 2023). The water consumption of AI should be considered within allocation arrangements to reach equitable solutions between users, particularly if current levels of water use are maintained or increase within water-scarce basins.

10.2 Data and information

Water resource systems cannot be effectively designed and operated unless adequate data and information are available concerning location, quantity, quality, temporal variability and demand (Stewart, 2015). Reliable hydrological data are needed to adaptively manage resources, to calibrate remotely sensed observations, and for modelling (Wilby, 2019). However, as highlighted throughout previous *United Nations World Water Development Reports,* there is a significant lack of data and information to inform the sustainable management of water. Government agencies tasked with resource monitoring and management often lack the capacity to collect data and generate information required to address water-related economic and social challenges (United Nations, 2023). This represents a significant challenge globally (UNESCO/UN-Water, 2020; Cantor et al., 2018; Stewart, 2015).

Where they do exist, data and information are often arranged in thematic or sectorial categories that have not significantly changed in recent decades (e.g. data and information are often compartmented for flood management, river discharge, water quality), without considering linkages (Wilby, 2019). Management challenges are exacerbated by climate change and associated hydro-meteorology variations, so that historical hydrological records can no longer be accurately used to predict future conditions (IPCC, 2022; Wagener et al., 2010; Milly et al., 2008). It is in societies' best interest that governments provide data though open-access platforms, without costs to users, and promote their dissemination (United Nations, 2023). Private companies should disclose relevant data and information concerning surface – and most notably sub-surface – water-related parameters to the authorities responsible for water management (United Nations, 2022). However, this requires relevant legislation at the national and potentially at the transboundary level.

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[49] Microsoft disclosed that its global water consumption spiked 34% from 2021 to 2022 (Microsoft, 2022), while Google reported a 20% growth in water use in the same period (Google, 2023).

[50] Except in Ireland where it can support 70 queries, owing to the cooler climate and relatively more water-efficient energy generation.

Box 10.1 Risks associated with cyber-attacks

The number of reported cyber-attacks on critical water infrastructure – including drinking water supply, wastewater and sewerage treatment, dams and canals – has increased in recent years (Tuptuk et al., 2021). These risks are expected to rise owing to the development and increasing uptake of cyber-physical water systems, that integrate computational and physical capabilities in order to control and monitor processes. In the past, water system security was achieved largely through physical isolation, limiting access to control components. However, with the emergence of the Internet of Things,[a] water systems are increasingly using a smart systems philosophy, incorporating analytics into industrial control systems to improve the sensing and control capacity (Bello et al., 2023; Tuptuk et al., 2021).

"Cyber-attacks could be launched remotely by employing command and control techniques to interrupt the system's performance and provide access to illegitimate parties to critical and confidential information. Moreover, in more severe cases, such attacks can even cause physical impairment to the system's structure. Furthermore, such attacks can hamper the water quality by changing the treatment systems or suppressing contamination warnings by affecting water quality sensors" (Bello et al., 2023, p. 2). The implications on society are potentially serious and multi-faceted. Cyber-attacks may affect services of critical infrastructure for drinking water, wastewater treatment and sewerage, agricultural production and food systems, energy generation, navigation, and disaster management (including floods and droughts) (Gleick, 2006; Amin et al., 2012; Copeland, 2010).

Governments are developing cyber-security plans to safeguard critical water infrastructure. To mitigate risks, personnel needs to be trained to assess and identify threats to water infrastructure (Bello et al., 2023; Moraitis et al., 2020; Hassanzadeh et al., 2020; Adepu and Mathur, 2016). Measures include regular cyber-security assessments and incident response plans, vigilant monitoring of water system treatment processes, along with access controls encryption, firewalls, anti-virus measures, back-ups and multi-factor authentication (Waterfall, 2023).

[a] The Internet of Things describes devices with sensors, processing ability, software and other technologies that connect and exchange data with other devices and systems over the Internet or other communications networks.

10.2.1 Data sources

Hydrological data can be obtained from a number of sources. These include in-situ measurements through monitoring networks operated by governments, and in some cases operated by other water users such as hydropower companies and private farms. Data are also obtained from model estimation and administrative collection (e.g. regulation data such as permits or census data) (Bureau of Meteorology, 2017). Data can also be generated by other sources, such as earth observations from satellites (Landerer and Swenson, 2012), sensor networks, citizen data and social media (UNESCO/UN-Water, 2020). Considerable amounts of data are still held in paper archives that have yet to be digitized, particularly government records in low-income countries; but also in high-income countries when it comes to historical hydro-meteorological variability (Burt and Hawkins, 2019). Once data are collected and analysed, they can be transformed into the information that is required to support management decisions and policy development processes.

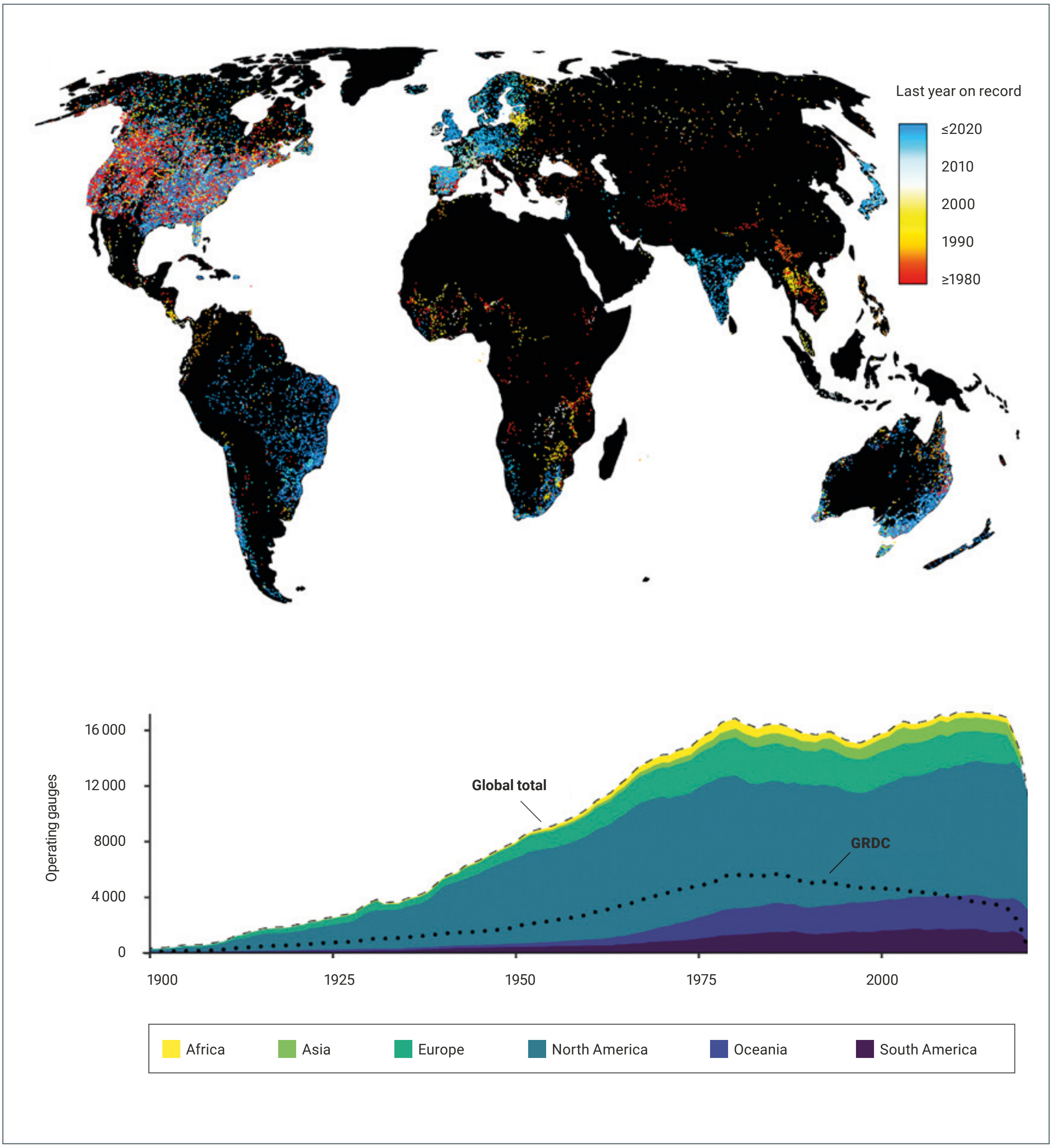
Global overview of data/information

Although some hydrological data are collected in nearly all major river basins around the world, their quality, distribution, availability and accessibility are highly variable; as well as the parameters monitored (WMO, 2022). In many low-income countries, hydrological data are largely inadequate. Data on water quantity and especially water quality remain sparse, due in large part to weak monitoring and reporting capacity. This is especially the case in many low-income countries in Africa and Asia (United Nations, 2023).

There are geographically large regions that lack ground measurements of fundamental water balance components (e.g. precipitation, evapotranspiration and changes in ice, lakes, wetlands, soil, or groundwater storage). Observational network densities are notably sparse in the Arctic, Sub-Saharan Africa (with the exception of South Africa), Central Asia, the Pacific Islands and South America (Wilby, 2019). A significant proportion of countries are characterized by very limited or inexistent groundwater monitoring networks, owing to the cost of development, operation and maintenance (IGRAC, 2020). Accurate data on soils are even more sparse, significantly limiting knowledge of how water soil content and storage impact agricultural productivity (Kendzior et al., 2022).

Gauging stations are distributed unevenly and sparsely across rivers globally (WMO, 2022). They do not capture the full extent of hydrological variability and anthropogenic influences (Krabbenhoft et al., 2022). The concentration of river gauging stations is significantly higher in Europe, North America and Oceania, relative to other regions (Figure 10.1). Gauging stations are distributed relatively sparsely within basins characterized by high annual river discharge, and rivers characterized by non-perennial flow regimes (Krabbenhoft et al., 2022). There is also a disconnect between the number of gauging stations in basins with a high seasonal variability in water availability, where it is important to monitor large inter-annual hydrological fluxes (Figure 10.2). There is a need to increase the number of gauging stations, particularly in under-represented basins and environmentally vulnerable areas, to capture the full extent of hydrological variability and anthropogenic influences.

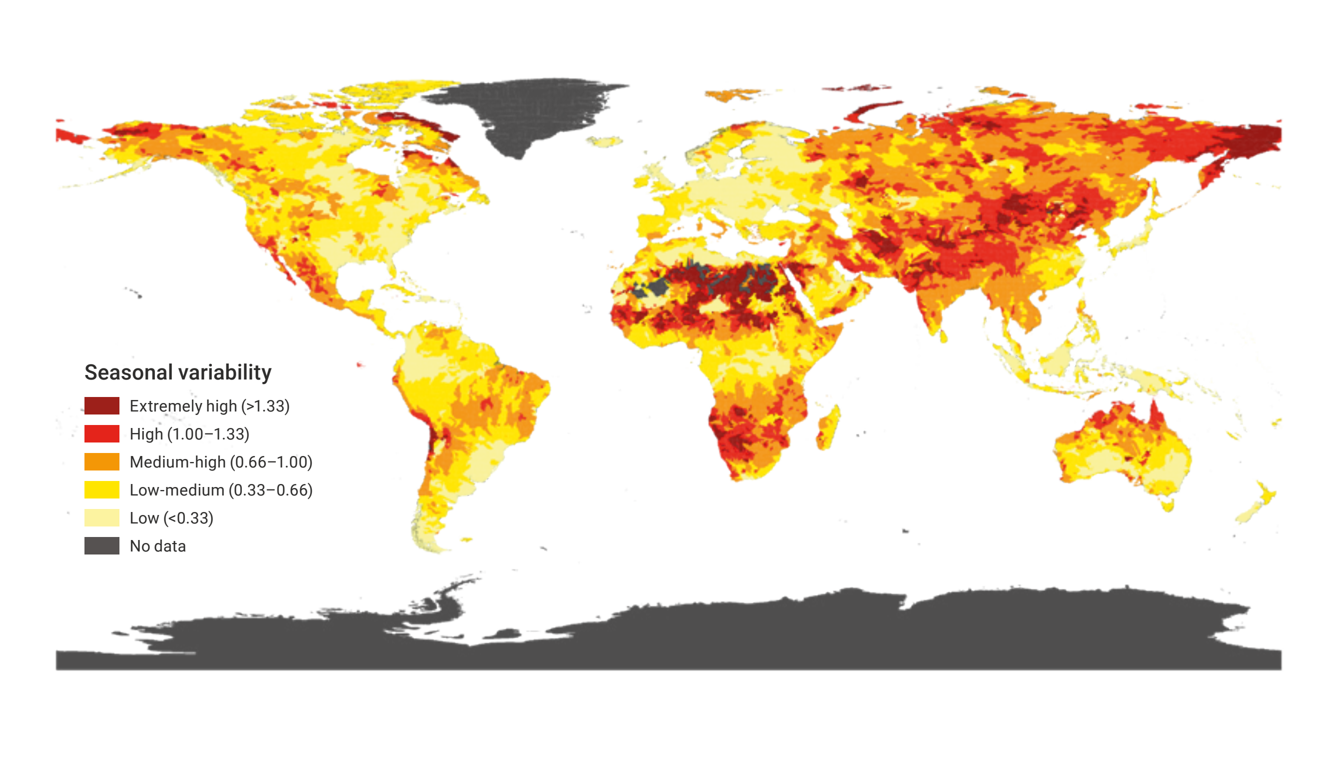
Figure 10.1 Global distribution of river gauging stations



Note: GRDC: Global Runoff Data Centre

Source: Adapted Riggs et al. (2023, fig. 2, p. 5). Licence CC BY 4.0.

Figure 10.2 Seasonal variability in water availability



Note: Seasonal variability measures the average within-year variability of available water supply, including both renewable surface and groundwater supplies. Higher values indicate wider variations of available supply within a year.

Source: WRI (2019). Licence CC BY 4.0.

Governments are developing cyber-security plans to safeguard critical water infrastructure

It is generally recognized that some of the most data-sparse regions are also the most vulnerable to hydroclimatic hazards (Wilby, 2019). High-altitude regions and fragile states are particularly under-monitored. For instance, high-altitude areas of the Hindu Kush Himalaya lack long-term observational data, and the available data suffer from large inconsistencies and from high inhomogeneity. This hinders understanding of system dynamics, and is *"a major barrier to designing context-specific interventions. This […] deters the addressing [of] challenges faced by mountain communities scale economics, access to infrastructure and resources, poverty levels and capability gaps, and also thwarts the large-scale replication of successful innovative demonstration/pilot projects that have been implemented in the region"* (Wester et al., 2019, pp. 168–169).

Even where data exist, records are often incomplete due to a lack of resources for personnel and equipment, or because they were lost due to inadequate data management practices, all of which are likely to be exacerbated under conditions of conflict. Furthermore, there is potential for erroneous issues due to site, instrument or observer changes; data may be corrupted at any point in the information flow or held in inaccessible formats (Wilby et al., 2017).

Responses to increase data collection and monitoring

As described by Wilby (2019), approaches to improve hydrological data collection and monitoring include the installation of equipment at strategically important places (known as sentinel locations) where hydrological variations are most likely to be detected (Fowler and Wilby, 2010). Evidence suggests that at high elevation, sites are warming more rapidly than the global average (Pepin et al., 2015), although this might only hold true up to 5,000 metres (Gao, et al., 2018). Hydrological assessments are urgently required in mountain regions; for instance, in the Himalaya Hindu Kush region, where mountain waters support the agricultural livelihoods, and water and energy requirements of over two billion people (Wester et al., 2019; Immerzeel et al., 2010).

Insights regarding data-sparse regions can be gained from several approaches. Basic hot-spot sensitivity analyses can identify communities most at risk, or inform the design of future hydro-meteorological network expansions (Wilby, 2019). Temporal data-scaling can extrapolate to sub-daily or sub-hourly precipitation levels for engineering design, in those situations where daily data exist (Courty et al., 2019). Alternatively, geostatistical techniques can be used to blend fragments of in-situ field data with remotely sensed information (Wilby and Yu, 2013), including proxies for hydrological variables (Najmaddin et al., 2017) to run hydrological models (Samaniego et al., 2011). Comprehensive assessments are needed to compare temporal and spatial variations in remotely sensed, model and ground-based hydro-meteorological data (Sun et al., 2018).

The development of high-resolution gravity-based methods could assess water variability at whole catchment scales, complementing or providing an alternative to river gauging techniques (Gouweleeuw et al., 2017). Remote sensing data may offer advantages to data-sharing between riparian countries not sharing data. Satellite imagery is impartial and can facilitate the incorporation of scientific data into decision-making. Remote sensing can aid in data collection, aggregation, monitoring, and information sharing (UNESCO/UN-Water, 2020; Wilby, 2019).

In-situ hydrological monitoring is often inadequate or absent in low-income countries

In-situ hydrological monitoring is often inadequate or absent in low-income countries. This is often due to insufficient technical capacity to operate and maintain equipment, particularly in remote locations, underpinned by low levels of funding to install, maintain and expand observation networks. In such cases, alternative approaches using low-cost, locally available technology (e.g. staff gauges and local observers), as well as existing capacity can be considered to collect and manage data. Citizen science is the involvement of citizens in scientific research and data gathering (Njue et al., 2019; Bonney et al., 2009). It represents an invaluable opportunity for both data collection and public participation in water-related projects (Hegarty et al., 2021). Over recent years, significant volumes of data and information on water science and management have been collected (Buytaert et al., 2014; Follett and Strezov, 2015). For instance, citizen science has been used to address the lack of water quality data in relation to SDG Indicator 6.3.2 (Quinlivan et al., 2020). Beyond producing data, citizen science is also recognized to have broader environmental, social, economic and political benefits (Hecker et al., 2018). These include strengthening participatory decision-making processes, local leadership and capacity development (Njue et al., 2019). Capacity-building through externally sourced training, in order to establish operation and maintenance of hydrological monitoring systems within data-sparse regions, has occurred historically in low-income countries with varied levels of success (Kirschke et al., 2020).

Data- and information-sharing

Forthcoming and transparent sharing of data and information is essential to promote effective water management. Transparent sharing includes data and metadata exchange, preferably within international standards; as well as open-access and open-source platforms and technologies.[51] However, the level of sharing varies significantly. Where data and information are considered sensitive (politically, economically or otherwise), sharing is often limited or non-existent. Data and information can be withheld or manipulated to serve one actor's interest over that of others. There can also be significant time lags between data collection and sharing, which could hamper operational decision-making.

Data-sharing can be limited between government and non-government actors. The private sector can limit data-sharing by declaring certain data as 'sensitive' (e.g. a security risk) to protect business interest, for instance, regarding infrastructure development, domestic water supply or agriculture projects. Lack of data-sharing is also reported by private

[51] Examples include the open-access platforms maintained by FAO, such as: FAOSTAT, AQUASTAT and WaPOR.

sector investment companies engaged in large-scale land acquisition in some countries (Dell'Angelo et al., 2018; Rulli et al., 2013; Mehta et al., 2012). Irrigation is often employed on land acquired to boost agricultural productivity, where data regarding the volume of increased water withdrawn are often not collected or shared (Rulli et al., 2013).

There is a growing call for the private sector to share data with governments and other actors when a project contract expires, albeit with security caveats considered (United Nations, 2022; Rulli et al., 2013). Furthermore, environmental impact assessment data within countries that are targeted for land acquisition are found to be lacking (Dell'Angelo et al., 2018). Data-sharing can also be limited between and within government ministries and departments, which are often characterized by ineffective communication between and within departments and ministries, along with competition for project funding.

Data- and information-sharing between riparians in transboundary basins is a long-standing issue of concern. In an assessment of transboundary river basins in Africa, Asia, Europe and North America, a study found data-sharing to be below the targets established at basin level and internationally[52] (IWMI, 2021). Although a reasonable proportion of river basins exchange some operational data, the breadth of exchange is often limited and irregular. Data-sharing is found more likely to take place if it responds to a particular operational need and serves practical uses, such as to minimize flood risks or to manage transboundary infrastructure (e.g. a reservoir) between riparians. This finding concurs with a recent study that found that the level of operational data-sharing at the international basin level is steadily increasing (WMO, 2022). Despite SDG Indicator 6.5.2 promoting transboundary cooperation for IWRM, there is no single global hydrological monitoring system. Instead, there is a proliferation of networks designed and operated by government and non-government actors for specific uses and at different spatial scales, covering a range of parameters and data types. There is a need to strengthen basin-level data exchange between actors, through river basin organizations or other relevant organizations.

10.3 Conclusions

It is widely acknowledged that more and better data and information are required to inform better water management decisions and policy development. Advances in science and technology have expanded the potential to monitor the hydrosphere in greater scope and detail, thereby increasing global and local knowledge bases.

The shared availability of credible and trusted data is fundamental to building confidence between water users; at the transboundary level, and within nations and river basins. This in turn helps effective and equitable water management, and promotes prosperity and peace within society. Where available and accessible, data and information empower the user to make an evidence-based informed decision. However, further advances in science and technology, delivering better and more data alone, will not necessarily improve decision-making and policy development (WaterAid, 2019; Kumpel et al., 2020). Actors and water users make decisions and develop policies based on a multitude of factors. These include but are by no means limited to socio-political, economic, technical, administrative and other factors. Whether in abundance, lacking or in secrecy, data and information can be used in a multitude of ways.

[52] The 1992 United Nations Economic Commission for Europe's *Convention on the Protection and Use of Transboundary Watercourses and International Lakes* (UNECE, 1992); and the *United Nations Convention on the Law of the Non-Navigational Uses of International Watercourses* (General Assembly of the United Nations, 1997).

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Water nurtures prosperity by meeting basic human needs, supporting health and livelihoods and economic development, underpinning food and energy security, and defending environmental integrity.

The impacts of climate change, geopolitical unrest, pandemics, mass migration, hyperinflation and other crises can exacerbate water access inequalities. In nearly all cases, the poorest and most vulnerable groups are those that suffer the greatest risks to their well-being and livelihoods.

The ***2024 edition*** of the ***United Nations World Water Development Report (UN WWDR)*** describes how developing and maintaining a secure and equitable water future underpins ***prosperity and peace*** for all. It calls attention to the complex and interlinked relationships between water, prosperity and peace, describing how progress in one dimension can have positive, often essential, repercussions on the others.

Water infrastructure promotes growth and prosperity by storing a reliable water supply and delivering it to economic sectors, including agriculture, industry, energy and the relevant business and service sectors that support billions of livelihoods. Similarly, safe, accessible and well-functioning drinking water and sanitation systems foster prosperity through quality of life, greater education opportunities and a healthy workforce.

Cooperation over water resources has generated positive and peaceful outcomes, ranging from participatory, community-led initiatives that have relieved local tensions, to dispute settlement and peacebuilding in post-conflict settings and transboundary watersheds. However, inequalities in the allocation of water resources, in access to water supply and sanitation services, and in the distribution of social, economic and environmental benefits can be counterproductive to peace and social stability.

The UN WWDR is UN-Water's flagship report on water and sanitation issues, focusing on a different theme each year. The report is published by UNESCO, on behalf of UN-Water and its production is coordinated by the UNESCO World Water Assessment Programme. The report gives insight on main trends concerning the state, use and management of freshwater and sanitation, based on work done by Members and Partners of UN-Water. Launched in conjunction with World Water Day, the report provides decision-makers with knowledge and tools to formulate and implement sustainable water policies. It also offers best practices and in-depth analyses to stimulate ideas and actions for better stewardship in the water sector and beyond.

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